

CITYTWIN · REQUIREMENTS

CityTwin requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/citytwin		
Source	docs/03-requirements.md @ 9033288		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

CityTwin requirements

These are first-pass requirements for the concept. Targets are proposals for review, awaiting Amish, and will be checked by calculation at TRL 3. The status column compares each target with the first-order estimates in the design precis (CTW-PRC-001). Four requirements are **not met** by the current concept: R10 (button response), R12 (display temperature range), R16 (kiosk cost) and R17 (cost with gateway). R1 is at risk.

Table 1. Requirements.

ID	Requirement	Target	Status (TRL 2 estimate)	Verification (TRL 3 or later)
R1	Ingest every node type in the smart city set	All 9: seven over LoRaWAN through TwinKit, PotholeLog by file upload over depot Wi-Fi, DockHub over LTE-M	At risk: LoRaWAN types met by design; PotholeLog and DockHub upload paths are not yet defined in those repos	Interface review with each sibling repo
R2	Capacity on one TwinKit gateway	50 nodes or more and 10,000 records per day or more	Met: reference neighborhood of 46 nodes sends about 4,900 records per day	Airtime and database load calculation
R3	Accept only counts and levels	Each node type has a payload schema; any field outside it is rejected and logged; no images or audio stored	Met by design	Schema review; fuzz test of the decoder
R4	Suppress small counts in public outputs	People and vehicle counts below a threshold (proposed 5 per hour) published as "fewer than 5"	Met by design; threshold awaiting Amish	Review of export code
R5	No movement traces in public outputs	PotholeLog published per road segment only, no vehicle tracks or times finer than one day; DockHub per dock per hour, no card IDs	Met by design	Review of export code
R6	Open data export	Hourly CSV and GeoJSON with a documented schema, under an open license, pushed to a public host	Met by design; OGC SensorThings API proposed	Sample export checked against the schema
R7	One map for all node types	One layer per node type, node health, and an "as of" timestamp on every view	Met by design	Review of dashboard mock-up
R8	Data freshness	Map and kiosk show data no older than 30 min in normal operation	Met, thin margin: about 25 min worst case for 15 min nodes	Timing calculation
R9	Kiosk readable day and night	Readable in direct sun and at night at 1.5 m; key text cap height 7 mm or more	Met by design (reflective e-paper, hood light); unverified	Legibility calculation, later mock-up
R10	Kiosk responds to a button	Selected layer shown within 5 s of a press	Not met: about 20 s full refresh of the color e-paper; the button LED ring acknowledges the press at once	Datasheet timing
R11	Accessible kiosk	Operable parts 380 to 1,220 mm above ground (ADA forward reach); same content on a web page that meets WCAG 2.1 AA, linked by QR code and short URL	Met by design for reach (buttons at about 1.12 m); no audio output on the kiosk	Design review; accessibility audit of the web page
R12	Kiosk survives the outdoor climate	Operate from -20 °C to +45 °C air temperature; head IP54, cabinet IP55	Not met: display rated 0 to 40 °C; needs a heater, a different panel or indoor siting	Thermal calculation of the head
R13	Kiosk stands up to wind	35 m/s gust with a factor of 1.5 without yield of post or anchors	Met by estimate: post bending stress about 12 MPa	Structural calculation

ID	Requirement	Target	Status (TRL 2 estimate)	Verification (TRL 3 or later)
R14	Electrical safety	Mains only inside the locked cabinet, behind a 30 mA RCBO and a surge protector; only 12 V SELV in the head; all metal earthed	Met by design	Design review by a qualified electrician
R15	Low running power	15 W average or less including the TwinKit gateway	Met: about 9 W	Power budget calculation
R16	Kiosk parts cost	\$600 or less for the kiosk (items 1 to 12, 14, 15)	Not met: about \$740	Priced BOM
R17	Cost with gateway	\$600 or less for kiosk and TwinKit gateway together (<code>budget_usd</code>)	Not met: about \$1,025	Priced BOM
R18	Honest in an outage	Screen keeps the last image with its timestamp when power fails; gateway rides through 2 h on its backup pack	Met by design (e-paper is bistable; TwinKit backup)	Design review
R19	Secure by default	No inbound connections from the internet to the gateway; open data pushed one way; no exposed ports or USB on the kiosk	Met by design	Security review

■ Assumptions

- Node types and reporting intervals follow the sibling READMEs: CurbCount, HeatMap Node and NoiseMap every 15 min; AirStreet every 5 min. CrossSafe (hourly activation summary), FloodGauge (15 min status plus alerts) and LoadZone (about 64 state changes and heartbeats per puck per day) are CityTwin assumptions to be confirmed with those projects.
- LoRaWAN packet loss of about 3 % for a single gateway in a dense street (estimate).
- 35 m/s design gust is an estimate for a first check; the local wind code governs any real installation.